

LUMING TANG

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EDUCATION

- Cornell University, Department of Computer Science** Jan.2019-Present
Ph.D. in Computer Science
Advisor: Prof. Bharath Hariharan
- Tsinghua University, Department of Physics** Sep.2014-Jun.2018
B.S. in Mathematics and Physics, Rank: 4/47
Double Major in Economics

PUBLICATIONS AND MANUSCRIPTS

- **Luming Tang**, Yexiang Xue, Di Chen, Carla P. Gomes. “Multi-Entity Dependence Learning with Rich Context via Conditional Variational Auto-encoder”. in *Thirty-Second AAAI Conference on Artificial Intelligence (AAAI-18)*.
- Zhongdao Wang*, **Luming Tang***, Xihui Liu, Zhuliang Yao, Shuai Yi, Jing Shao, Junjie Yan, Shengjin Wang, Hongsheng Li, Xiaogang Wang. “Orientation Invariant Feature Embedding and Spatial Temporal Regularization for Vehicle Re-identification”. in *IEEE International Conference on Computer Vision (ICCV 2017)*. (* indicates equal contribution)
- **Luming Tang**, Boyang Deng, Haiyu Zhao, Shuai Yi. “Hierarchical Deep Recurrent Architecture for Video Understanding”. in *CVPR 2017 Workshop on Youtube-8M Large-Scale Video Understanding*.

RESEARCH EXPERIENCE

- Computer Vision Lab, Cornell University** Jun.2019-Present
Advisor: Bharath Hariharan, Assistant Professor
- Revisited pose-normalization representation for fine-grained few-shot learning. Paper has been submitted to CVPR 2020.
- Visual Computing Group, Microsoft Research Asia** Sep.2018-Dec.2018
Mentor: David Wipf, Principal Researcher
- Analyzed the regularization balance of Autoencoder-structured models in general and VAEs in particular, including local minima properties and necessary energy function characteristics. This leads to useful practical prescriptions and the first demonstration of high-quality, diverse generation results from Autoencoder-structured, non-adversarial training on high-resolution images.
- Department of Computer Science, Cornell University** Jun.2017-Sep.2017
Advisor: Carla Gomes, Professor
- Proposed a variational auto-encoder based algorithm to model structured multi-entity distribution, achieving better performance on two real-world applications compared to previous state-of-the-art approximate inference based methods and classifier chains. The first-author paper was accepted by AAAI.
- Research Department, SenseTime Group Limited** Dec.2016-Jun.2017
Mentor: Shuai Yi, Research Director
- Explored vehicle re-identification problem and proposed orientation invariant embedding together with spatial-temporal regularization. The model doubled the matching accuracy of previous methods on four publicly available datasets. The co-first-author paper was accepted by ICCV.

- Researched video understanding problem and proposed the deep recurrent architecture, achieving significant performance in Kaggle Youtube-8M challenge. The first-author paper was accepted by CVPR Video Understanding Workshop.

Natural Language Processing Lab, Tsinghua University

Sep.2016-Jun.2018

Advisor: Zhiyuan Liu, Associate Professor

- Constructed an open-source framework OpenNRE for neural relation extraction with labmates. The code was released on THUNLP Github homepage, receiving over 1300 stars and 400 folks.

TEACHING

CS4787 Principles of Large-Scale Machine Learning, Teaching Assistant

Spring 2019

CS2110 OOP and Data Structures, Teaching Assistant

Summer 2019

CS6670 Graduate Computer Vision, Teaching Assistant

Fall 2019